YTL*AI LABS

BUSINESS PROPOSAL

UMHackathon 2026

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1. Executive Summary:

Supply chain disruptions within the automotive manufacturing sector are not just logistical annoyances; they are catastrophic financial liabilities. Automotive SMEs operate interdependent lines where a single delayed component can trigger OEM downtime penalties ranging from RM 45,000 to RM 380,000 per day. Currently, supply chain managers bridge unstructured disruption data (e.g., supplier emails) and structured ERP databases using manual intuition, leading to slow response times and panic-driven over-ordering known as the Bullwhip Effect.

ChainLogic AI is an agentic decision intelligence system engineered specifically to solve this data fragmentation. Powered by the Ilmu GLM, it autonomously ingests unstructured supplier emails, extracts affected SKUs, fuses this with live SQLite ERP data, and generates mathematically optimized financial trade-offs. Targeting a Total Addressable Market (TAM) of RM 40 Billion in the global automotive supply chain software sector, ChainLogic AI cuts disruption decision latency from over 12 hours to under 30 seconds, averting ruinous penalties and saving SMEs an average of RM 35,000+ per incident.

2. Product Overview / Solution

Key Features: *Highlight key features of your product*

- ***Agentic SKU Extraction:*** *Ilmu GLM reliably reads free-form supplier emails and returns only the affected part number, eliminating manual data entry.*
- ***Live ERP Fusion:*** *Fetches real-time inventory levels, unit costs, and downtime penalty rates dynamically from the SQLite database to ground the AI's reasoning context.*
- ***Financial Trade-Off Analysis:*** *Generates 2–3 named recovery options (e.g., Expedite Freight vs. Reschedule) with precise net financial impacts and formulaic computation breakdowns.*
- ***Safe-State Short-Circuit:*** *If current stock levels already meet production requirements, the system bypasses the AI reasoning phase entirely, instantly returning a "SAFE" status. This prevents alert fatigue and reduces API token costs by an estimated 70–80%.*
- ***Closed-Loop ERP Execution:*** *Upon user approval, the backend autonomously updates production and inventory tables without requiring human re-entry, closing the decision loop instantly.*

Technology Used: *Highlight the technology used in your application*

- ***Decoupled Micro-Architecture:*** *Our Next.js frontend and Python FastAPI backend operate on a completely decoupled architecture. This means ChainLogic AI can sit on top of legacy ERP systems as a lightweight, non-intrusive 'intelligence layer' without requiring a massive, risky system overhaul.*
- ***Database Abstraction & Scalability:*** *We utilize a fully normalized SQLite relational database for the zero-config prototype, entirely abstracted via the SQLAlchemy ORM. This guarantees enterprise scalability, allowing a seamless, zero-code transition to an AWS RDS PostgreSQL cluster simply by swapping environment variables.*

- ***CI/CD & Serverless PaaS Automation:*** *The application is fully containerized using Docker. The deployment pipeline utilizes native GitHub Actions to push decoupled frontend components to Vercel and backend container workloads to Render, ensuring zero-downtime rolling updates and version control integrity.*
- ***3-Stage AI High-Availability Engine:*** *To ensure absolute enterprise reliability during API outages or rate limits, we engineered a deterministic 3-stage LLM cascade protocol:*

  Layer 1: Primary Reasoning Engine (Ilmu GLM-5.1).
  Layer 2: Secondary Failover (Groq / OpenAI GPT-OSS-120B).
  Layer 3: Global Open-Source Safety Net (OpenRouter Fallback Models).

- ***High-Speed Regex Pre-Processing:*** *We implemented a High-Speed Regex Pipeline that scans incoming unstructured disruptions. It identifies and extracts affected SKUs in 0.01 seconds, bypassing the AI latency entirely for simple entity extraction to drastically reduce compute overhead.*
- ***Token-Optimized 'Safe-State' Logic:*** *Before triggering an expensive LLM crisis analysis, our Python backend cross-references the extracted SKU with live SQLite inventory. If stock levels are mathematically 'Safe' or penalty risks are negligible, it short-circuits the process, bypassing the AI reasoning phase entirely to reduce token costs by an estimated 70%.*
- ***Dynamic Unit Economics ROI Tracking (Live):*** *We implemented real-time interception of the AI provider's JSON API payload to track exact usage.total_tokens. The UI dynamically calculates the true USD cost of the autonomous decision (e.g., $0.02) and graphs it against the live ERP downtime penalty saved (e.g., $45,000), providing executives with real-time, undeniable ROI unit economics for every disruption mitigated*

3. Market Research & Market Dynamics

3.1 Problem Landscape

SME supply chain managers suffer from critical data fragmentation. Disruption notices arrive in messy, unstructured formats (emails, PDFs), while the solution data (inventory, contracts, lead times) is locked in rigid ERP databases. During a crisis, human operators default to intuition-based panic ordering, amplifying the Bullwhip Effect and tying up working capital in excess inventory. According to Supply Chain Dive data, the average manufacturing supply chain disruption costs RM 2.8 Million globally per day, making this an urgent financial bleeding point for SMEs.

3.2 Market Niche

ChainLogic AI directly targets Tier-2 and Tier-3 local automotive manufacturing SMEs in Malaysia and Southeast Asia. These organizations require the deep analytical quantitative power of Tier-1 OEMs but cannot afford dedicated in-house data science teams.

3.3 Market Sizing

Estimations are based on typical SaaS penetration models and verified by Grand View Research (2023), automotive software data.

Market Tier	Definition	Estimated Value (2026)
TAM	Global Automotive Supply Chain Software Market	RM 40 Billion
SAM	Southeast Asian Automotive SME Digitalization Market	RM 3.5 Billion
SOM	Malaysian Tier-2/3 Automotive Suppliers (24 months)	RM 45 Million

Formula & Source :

TAM - Total Global Automotive Software Spend x Supply Chain Segment Share. (Validated by Grand View Research 2023, predicting a $12.3B industry size)

SAM - TAM x SEA Regional Adoption Rate (approx. 8.75%)

SOM - SAM x Expected SME Capture Rate in Malaysia (1.28%)

3.4 Market Dynamics & Placement:

[There were few key dynamic that shape ChainLogic AI entry to this market]:

- **Data Fragmentation:** Critical information exists in incompatible formats, creating a massive whitespace for multimodal AI systems capable of fusing text with SQL.
- **High-Penalty Environment:** With OEM downtime penalties reaching up to RM 85,000 per project per day, rapid, mathematically sound recovery strategies are highly lucrative.
- **SME Disadvantage:** Local manufacturers lack enterprise decision-support tools, leaving them uniquely vulnerable to ruinous line stoppages and margin compression.

4. Value Proposition & Competitive Advantage

ChainLogic AI acts as an always-on digital quantitative analyst, redefining how SMEs handle force majeure events.

- ***High Cost Efficiency:*** *The dual-call agentic pipeline consumes roughly 2,360 tokens per session. Formula: Total Tokens x Ilmu Pricing = Cost per Analysis. This yields a cost of less than RM 0.05 per crisis analysis, allowing for highly disruptive SaaS pricing margins*
- ***System of Action (Closed-Loop):*** *Unlike passive dashboards, ChainLogic AI is a closed-loop system. It executes the optimal recovery plan directly into the ERP, ensuring operational velocity and a 99% reduction in decision latency.*
- ***Zero-Friction Intelligence Layer:*** *Legacy ERP implementations take 6-12 months. ChainLogic AI is a lightweight intelligence layer that sits on top of existing databases. Our decoupled API architecture allows for deployment and ROI generation in under 14 days without ripping out legacy systems.*
- ***Anti-Hallucination Execution (The "System of Action")****: Standard LLM chatbots are read-only and prone to hallucinating financial numbers. ChainLogic AI uses the Z.AI GLM strictly for natural language extraction (reading the supplier email). The actual financial trade-off mathematics and ERP database updates are hard-computed by our Python backend using verified SQLite data, completely eliminating hallucination risk.*

5. Competitor Analysis (Porter's Five Forces)

Force	***Intensity***	***Assessment***
Competitive Rivalry	*Medium*	• *Legacy ERPs (SAP, Oracle) offer rigid supply chain modules but cannot natively interpret unstructured vendor emails.*
Threat of New Entrants	*Medium*	• *General-purpose AI wrappers are cheap, but ChainLogic's strict anti-hallucination logic and closed-loop database write access form a robust technical moat.*
Bargaining Power of Buyers	*High*	• *SMEs are cost-conscious; however, our sub-RM 0.05 API cost per session allows for highly competitive, value-based pricing that mitigates this.*
Bargaining Power of Suppliers	*Low*	• *We currently leverage Ilmu and Groq, but our decoupled API architecture allows for total model agnosticism if token pricing changes.*

Threat of Substitutes	*Low*	• *The primary substitute is the status quo: manual spreadsheets and intuition-based ordering, which consistently fails under time constraints.*

Competitor Comparison Table:

Competitor	*Strength*	*Weakness*	*ChainLogic AI's Advantage*
Legacy ERPs (SAP, Oracle)	Deep structural integration and master data authority.	Inflexible; cannot autonomously ingest or reason over unstructured supplier emails.	Agentic extraction and reasoning are applied as a lightweight, intelligent layer on top of existing data.
Standard LLM Chatbots (ChatGPT/Claude)	High conversational fluidity and ease of use.	Prone to severe financial hallucinations and cannot execute actions back to an ERP database.	Anti-hallucination hard-overwrites and closed-loop database execution ensure 100% data integrity
Manual Spreadsheets & Human Intuition	Zero software cost.	12+ hour decision latency, prone to panic-ordering and the Bullwhip Effect.	Sub-30-second automated execution backed by verifiable mathematics.

ChainLogic AI SWOT Analysis

6. Business Model Canvas

6.1. Customer Segmentation

1. ***Supply Chain Managers:*** *Seek fast, quantified decision support to avoid manual spreadsheet work during a crisis.*
2. ***Operations Directors:*** *Require high-level visibility of financial exposure, risk in dollar terms, and auditable action logs.*
3. ***SME Founders/CEOs:*** *Need to authorize strategic decisions quickly based on clear financial ROI and automated recommendations.*

6.2. Revenue Stream

Primary: B2B SaaS Subscription (Tiered monthly licensing)

- Tier 1 (Standard SME): RM 1,000/month per SME (approx. 12,000/year)
- Tier 2 (Enterprise License): RM 3,000/month per SME. Includes Dedicated VPC Hosting, unlimited API calls, and custom LLM prompt fine-tuning on proprietary vendor contracts.

Secondary: Implementation and Integration Fees

- Pricing Strategy: One-time Integration Fee of RM 3,000 per new client for connecting our intelligence layer to their existing database.

Formula used: Total Revenue = (Active Users x Monthly Subscription Fee x 12) + (New Users x Integration Fee)

Unit Economics (Per SME):

1. ***LTV (Lifetime Value): RM 36,000.***
- *Formula: Annual Revenue (RM 12,000) × Estimated Retention (3 Years).*
2. ***CAC (Customer Acquisition Cost): RM 5,000.***
- *Formula: (Partner Commission + Marketing Spend) / Total Conversions.*
3. ***LTV: CAC Ratio: 7.2:1.***
- *Analysis: This exceeds the 3:1 VC industry standard, indicating a highly efficient and scalable business model*

6.3. Cost Structure

Cost Drivers	Monthly Estimate (Year 1)	Cost Type	Details
Cloud & Infra	RM 850	Fixed Cost	FastAPI hosting (AWS), Vercel, and SQLite maintenance.
LLM Tokens	RM 250	Variable Cost	Based on ~5,000 sessions/month at ~RM 0.05 per session.
R&D / Security	RM 3,000	Fixed	Core engineering maintenance and security compliance.
GTM & Sales	RM 5,000	Overhead	Partner referral fees, direct marketing, and SME onboarding.
Total OpEx	RM 9,100		Annual OpEx: RM 109,200

6.4. Key Partners

Internal / Tech	Ilmu AI (Core reasoning engine via ilmu-glm-5.1) & Groq (High-speed GPT-OSS reasoning)
External / Market:	Regional IT consultancies specializing in SME ERP modernization.

6.5 Risk Assessment

Risk Type	**Impact**	**Likelihood**	**Mitigation**
SKU Extraction Failure	High	Medium	Implemented a fallback regex pattern to extract SKUs if the GLM fails or encounters unusual text formatting.
Hallucinated Financials	High	Low	The Python backend intercepts the GLM response and overrides baseline impacts using verified SQLite data.
Triggering Bullwhip Effect	Medium	Medium	The system prompt explicitly instructs the AI to calculate trade-offs that mitigate the bullwhip effect.
API Outage/ AI Latency	High	Low	Engineered a proprietary 3-Stage Fallback Architecture (Ilmu → Groq → OpenRouter) ensuring the ERP never crashes.

7. Implementation & Financials

7.1 Go-to-Market Strategy

ChainLogic AI will utilize a Channel Partner-First strategy to scale. We will partner with regional IT Consultancies and ERP Integrators who currently manage the digital infrastructure for Malaysian automotive SMEs

- **Partnership Model:** We offer a 20% recurring revenue share to consultants who bundle ChainLogic AI as an "Intelligence Layer" for their existing ERP clients.
- **Advantage:** This leverages established trust networks, reduces our sales cycle from 6 months to 4 weeks, and maintains our low RM 5,000 CAC.
- **Initial Wedge:** A 3-month Lightweight Integration Pilot to prove the RM 35,000+ incident savings before converting to a full annual license.

7.2 Implementation Plan

Phase	Action	Timeline	Target
Soft Launch	Deploy to a single pilot automotive SME; monitor SKU extraction and ERP write correctness	Month 1-4	1 Pilot Enterprise
Expansion Phase	Expand to 5 SME pilots; introduce A/B testing on extraction accuracy.	4-12 Months	5 Active Enterprises
Optimization	Full regional rollout; begin developing multi-agent negotiation features.	Month 12-18	Reach operational breakeven
TOTAL	Estimated Seed Investment	18 Months	Significant Market Share

7.3 Revenue Projections (3-Year)

Assumptions: RM 1,000/month SaaS fee; RM 3,000 one-time setup fee per client.

Metric	*Year 1*	*Year 2*	*Year 3*
Active SME Clients	*15*	*85*	*220*
Avg. Monthly API Calls	*4,500*	*25,500*	*66,000*
Subscription Revenue	*RM 180,000*	*RM1,020,000*	*RM 2,640,000*
Integration/Setup Fees	*RM 45,000*	*RM 255,000*	*RM 660,000*
Total Revenue	*RM 225,000*	*RM 1,275,000*	*RM 3,300,000*

- *Formulas used: Subscription Revenue = Active Clients x RM 1,000 x 12 months.*
- *Setup Fees = New Clients Added in Year x RM 3,000.*

7.4 Key Financial Milestones

- Month 3: Complete deployment with first pilot SME, proving the RM 35,000+ incident savings.
- Month 9: Reach operational breakeven as subscription revenues eclipse fixed costs.
- Month 18: Hit RM 4.5M ARR run rate, triggering Series A funding to expand into predictive forecasting.